Supplementary Information for *When Do Activation-Patching Effects Add Up? Evidence from Controlled CNNs*

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S1 Architecture and training specification

For each task and renderer block, the data generator draws 128 training, 64 validation, and 64 test contexts from independent deterministic streams. Each context specifies rendering conditions such as angle, scale, and noise. Rendering all four classes in every context gives 512/256/256 images in the three splits. Models are trained for 200 epochs with Adam, learning rate 0.003 and batch size 128. Retention keeps the first-layer penalty at $\lambda = 1$ throughout training. Release uses the same penalty through epoch 10, decreases it linearly to zero at epoch 80, and leaves it at zero thereafter. In the architecture study, paired retention and release runs use the same model initialization and paired minibatch-order streams. Pairing is within each renderer block, task, architecture, and initialization replicate.

All architectures use a bias-free $1 \to 16$, 9×9 first convolution with same padding and ReLU. Patching is applied to the post-ReLU output of this layer. GMP and GAP denote global max and global average pooling. Plain depth-2 models add one downstream 3×3 convolution; plain depth-4 models add three. Width-64 variants expand the first downstream convolution from 16 to 64 channels and remain width 64 thereafter. Residual depth-4 models remain width 16 and add the third downstream convolution to the original first-layer activation before the final ReLU. BatchNorm, when present, occurs only after the patching location, following each downstream convolution. A learned linear classifier maps the pooled representation to four classes.

The architecture constructor and training routines are provided in the separate reproducibility archive's architecture-comparison code group. The frozen design and

Table S1 Complete 16-model architecture grid. Depth counts the shared first convolution.

Architecture	Depth	Width	Pool	Residual	BatchNorm	Parameters
`tiny_gmp`	1	16	GMP	no	no	1,364
`tiny_gap`	1	16	GAP	no	no	1,364
`plain2_w16_gmp`	2	16	GMP	no	no	3,684
`plain2_w16_gap`	2	16	GAP	no	no	3,684
`plain4_w16_gmp`	4	16	GMP	no	no	8,324
`plain4_w16_gap`	4	16	GAP	no	no	8,324
`plain2_w64_gmp`	2	64	GMP	no	no	10,836
`plain2_w64_gap`	2	64	GAP	no	no	10,836
`plain4_w64_gmp`	4	64	GMP	no	no	84,692
`plain4_w64_gap`	4	64	GAP	no	no	84,692
`res4_w16_gmp`	4	16	GMP	yes	no	8,324
`res4_w16_gap`	4	16	GAP	yes	no	8,324
`bn2_w16_gmp`	2	16	GMP	no	yes	3,700
`bn2_w16_gap`	2	16	GAP	no	yes	3,700
`bn4_w16_gmp`	4	16	GMP	no	yes	8,372
`bn4_w16_gap`	4	16	GAP	no	yes	8,372

parameter counts are recorded in its architecture data group. The archived code is supplied for inspection; a complete retraining run additionally requires the raw experiment store.

S2 Training dynamics and filter structure

The supplementary plots document the behavior of the first-layer templates throughout training and the corresponding patching-score trajectories. Figures S1 and S3 aggregate multiple saved runs across renderer blocks and report *mean* ± *standard deviation*, not confidence intervals. Figure S2 plots the *archived interval bounds* for each aggregate patching estimate; these bands must not be interpreted as run-to-run standard deviations. Figures S4 and S5 show only the specified illustrative example (`two_concepts`, TinyCNN, renderer block 2000), at epochs 0 and 200. These visualizations do not independently establish functional importance.

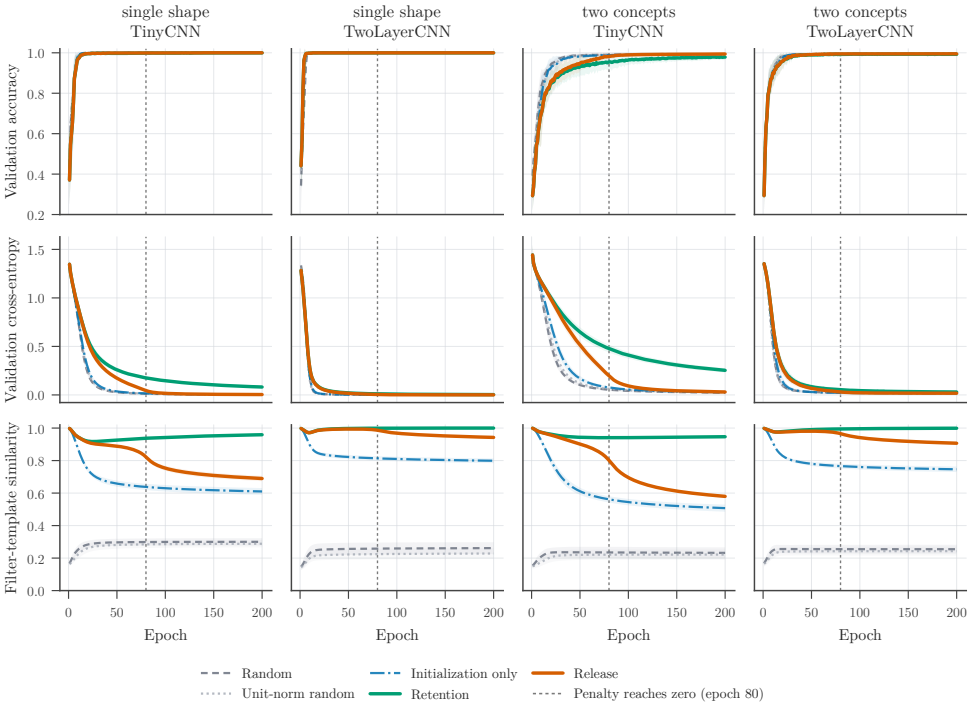


Fig. S1 Training dynamics for both tasks and two architectures. Curves show mean validation accuracy, validation cross-entropy, and filter–template alignment by training condition; shaded regions are ± 1 standard deviation across saved runs. The vertical line marks epoch 80, when the default release penalty reaches zero.

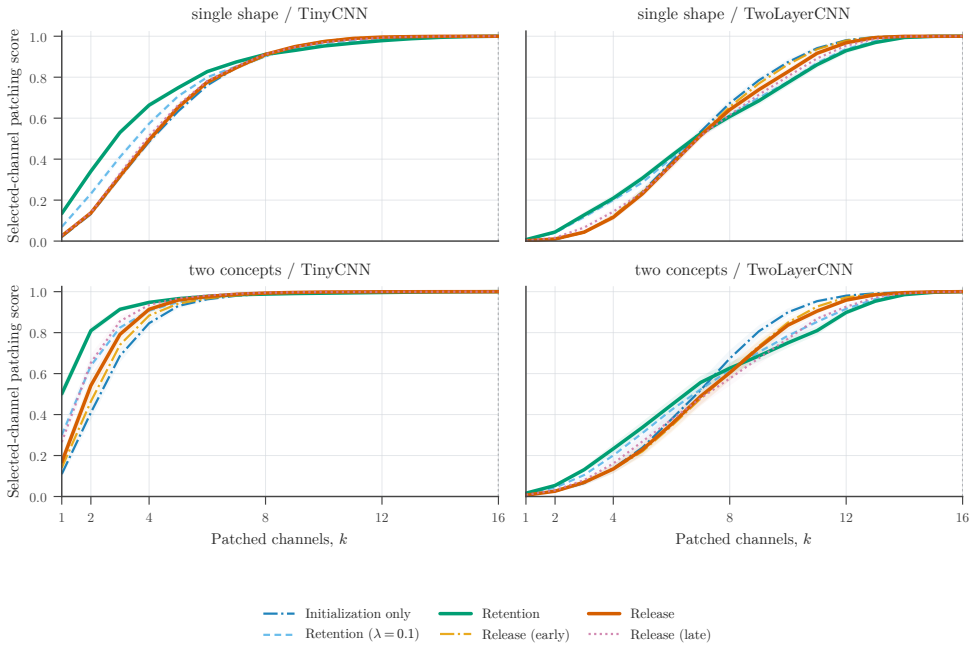


Fig. S2 First-layer selected-channel patching scores as a function of channel count k for the original training schedules, both tasks, and two architectures. Shaded regions use the archived `ci_low` and `ci_high` interval bounds in the aggregate curve table (not run-to-run standard deviation). Retention and release designate the constant-penalty and annealed-penalty conditions.

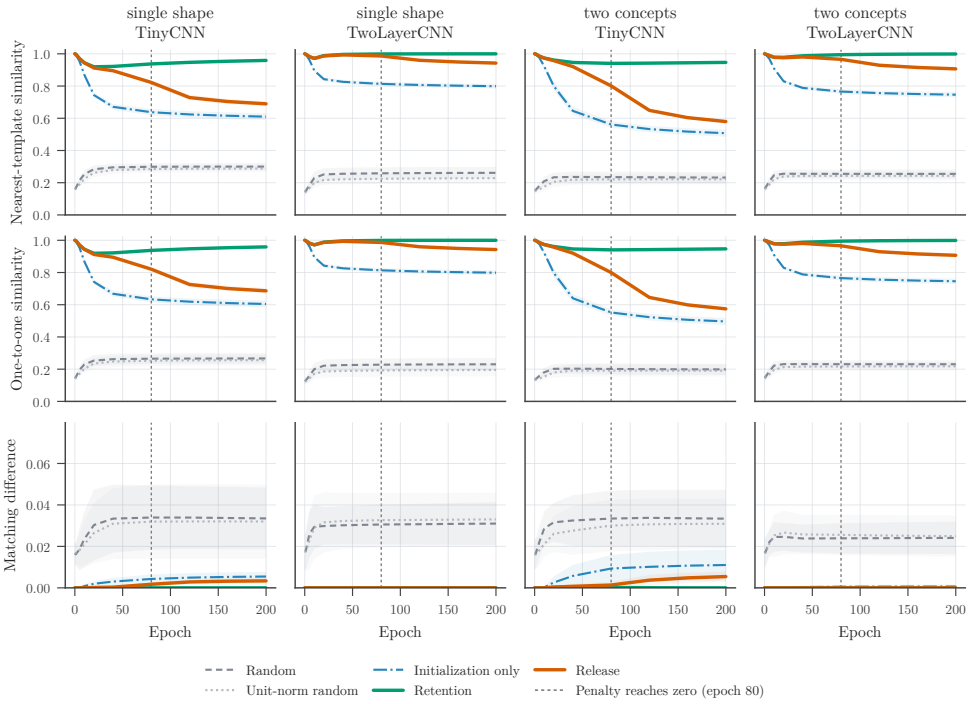


Fig. S3 Filter–template matching over training, aggregated across the saved runs of each task and architecture. Curves and shading show mean ±1 standard deviation. Nearest-template similarity allows repeated use of a template; one-to-one similarity uses maximum-sum bipartite matching (Hungarian assignment), assigning each learned kernel to a distinct template. Matching difference is their signed difference. Each spatial kernel is centered and normalized to unit norm before computing signed cosine similarity.

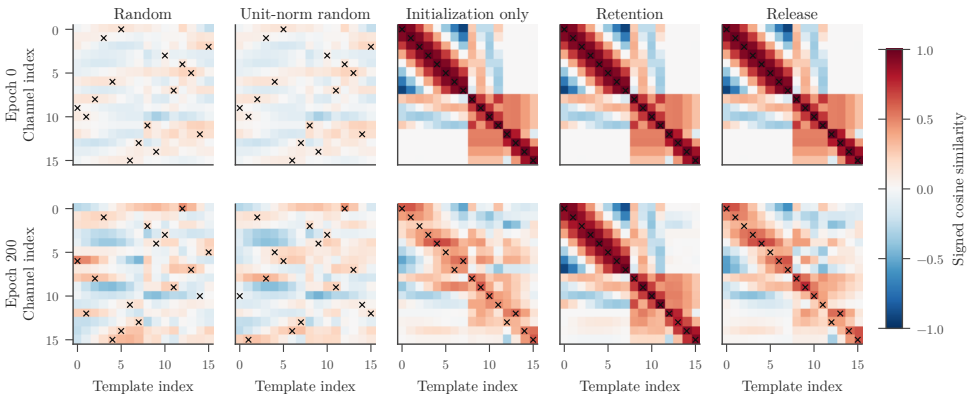


Fig. S4 Illustrative 16×16 signed cosine-similarity matrices for `two_concepts`/TinyCNN, block 2000, at epochs 0 and 200. Rows index learned filters and columns index predefined templates. Black crosses denote the maximum-sum one-to-one (Hungarian) matches, not independent nearest neighbors. All panels use the same $[-1, 1]$ color range.

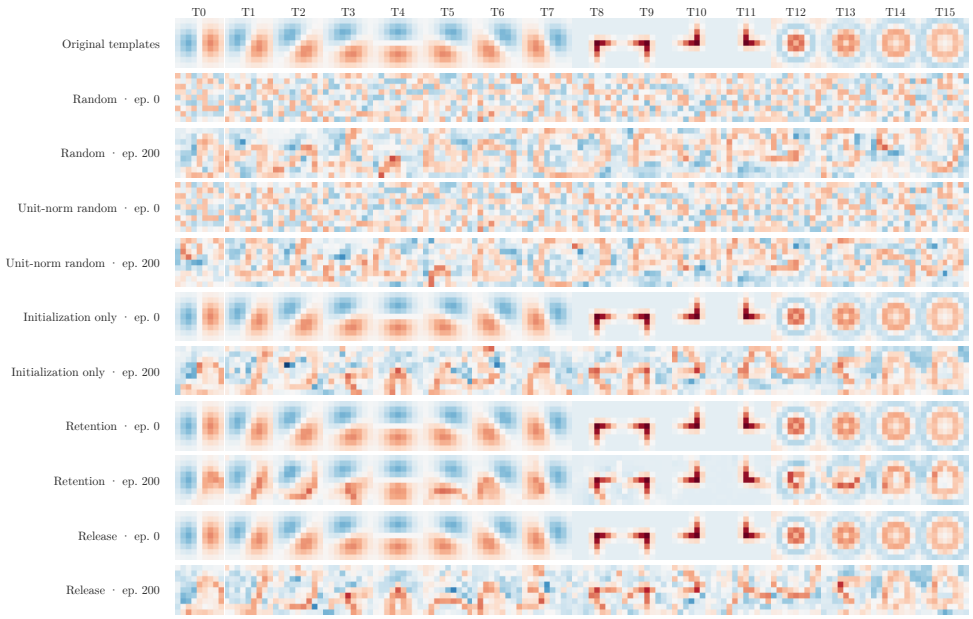


Fig. S5 Complete filter gallery for the same `two_concepts`/TinyCNN example, renderer block 2000. Original templates and all five original training conditions at epochs 0 and 200 are displayed. Kernels are ordered by the one-to-one maximum-similarity template assignment; each is spatially centered and normalized to unit norm. All rows share the same symmetric color limit, including the compact three-row comparison in the main article. The plot shows weight structure only, not patching effects.

S3 Intervention-size and metric robustness

For a patching metric M, the main intervention-size contrast is

$$B^M = \frac{\Delta^M(4) + \Delta^M(8)}{2} - \frac{\Delta^M(1) + \Delta^M(2)}{2}, \tag{S1}$$

where $\Delta^M(k)$ is release minus retention at intervention size k. The initial pre-specified test uses 20 held-out renderer blocks and the primary `two_concepts`/TinyCNN setting.

Table S2 Primary intervention-size contrast under the initial metric and two alternative evaluation metrics. Intervals are paired 95% intervals over the 20 held-out renderer blocks.

Metric	B	95% CI
Probability fidelity	+0.33098	[0.27368, 0.38827]
Centered-logit fidelity	+0.17888	[0.12727, 0.23049]
Probability-error reduction	+0.89441	[0.78630, 1.00252]

The alternative metrics reuse the frozen checkpoints and validation-derived channel rankings. Their metric definitions and analysis plan were fixed before evaluating the alternative-metric outcomes. Full pointwise curves and block-level values are provided in the reproducibility archive's metric-sensitivity data group.

S4 Architecture robustness

The architecture study uses 100 new renderer blocks and four paired initialization replicates per block. It covers the 16 architectures in Table S1, two tasks, and both training regimes. Initialization replicates are averaged within renderer block before inference.

Table S3 Repeated-measures tests of differences across architectures on the primary `two_concepts` task.

Metric	F	df	partial η^2	Holm p
Centered-logit fidelity	77.09	15, 1485	0.438	2.40×10^{-173}
Probability-error reduction	656.70	15, 1485	0.869	$< 10^{-300}$

The architecture-specific estimates plotted in the main paper are in the accompanying archive's architecture data group, together with the training design, factor contrasts, input-effect diagnostics, and omnibus results. Classification accuracy is not

matched across architectures. These comparisons describe the trained networks in this controlled grid.

After the pre-specified architecture results were known, a post-hoc analysis checked whether differences across architectures remained after sphericity correction, within-block permutation tests, Friedman tests, and random channel orders. The original architecture test was pre-specified; these additional robustness checks were post-hoc.

Table S4 Post-hoc checks of differences across architectures on `two_concepts`. Permutation values are Holm adjusted across the two metrics.

Metric	GG ϵ	GG p	Permutation Holm p	Friedman p
Centered-logit fidelity	0.1861	1.56×10^{-34}	2.0×10^{-5}	3.52×10^{-146}
Probability-error reduction	0.5973	$< 10^{-300}$	2.0×10^{-5}	7.79×10^{-200}

S5 Pixel-permuted spatial control

The spatial-control experiment compares the original structured filter bank with a bank obtained by applying one common permutation to the 81 spatial coordinates of every filter. This transformation preserves row norms, the complete Gram matrix, numerical rank, singular values, and the coefficients in each filter, including how often each value occurs. It destroys the original two-dimensional arrangement. The primary analysis uses 100 new renderer blocks and four architectures.

Table S5 Difference between the structured and pixel-permuted banks, averaged across architectures, for channels selected on the validation set in `two_concepts`. Difference intervals are paired Student-t 95% intervals from the primary spatial-control analysis; equivalence uses the pre-specified 90% TOST interval and symmetric margin.

Metric	Mean difference	95% CI	90% CI	Margin
Centered-logit fidelity	+0.05743	[0.05116, 0.06369]	[0.05219, 0.06266]	±0.01704
Probability-error reduction	+0.04563	[0.03728, 0.05397]	[0.03864, 0.05261]	±0.06594

The centered-logit interval lies outside its equivalence region. The probability-error 90% interval lies inside its wider study-specific equivalence margin even though its 95% difference interval excludes zero.

The negative difference in `plain2_w16_gmp` shows that the original bank does not increase the contrast in every architecture. The spatial effect depends on architecture and metric.

Table S6 Structured-minus-pixel-permuted B shown by architecture on `two_concepts`.

Architecture	Centered-logit difference	Probability-error difference
`tiny_gmp`	+0.10624	+0.19953
`tiny_gap`	+0.07690	+0.04308
`plain2_w16_gmp`	-0.04453	-0.08067
`plain2_w16_gap`	+0.09109	+0.02057

S6 Reconstructing joint outputs from individual patches

We re-evaluate saved checkpoints by patching one first-layer channel at a time. We then reconstruct the jointly patched output by summing these individual centered-logit changes against the same base run. We evaluate all $k = 0, \ldots, 16$ and use $k = 1, 2, 4, 8$ to compute B. The protocol, implementation manifest, numerical unit tests, and analysis code are included in the accompanying archive's additivity study group (see the source-path manifest for original identifiers).

Table S7 Primary examples from the reconstruction using individual patches on `two_concepts`. The BN2-GAP paired interval is computed over the same 100 renderer blocks.

Architecture / metric	Observed B	Additive B	Observed − additive
`tiny_gmp`, centered logits	+0.14558	+0.14558	numerical precision
`tiny_gmp`, probability error	+0.83151	+0.83151	numerical precision
`bn2_w16_gap`, centered logits	-0.13817	+0.04177	-0.17994

For `bn2_w16_gap`, the paired centered-logit difference is -0.17994 with 95% Student-t interval $[-0.20082, -0.15906]$. In the analytic `tiny_gmp` control, centered-logit residual energy remains at numerical precision: 8.41×10^{-14} for release and 1.45×10^{-13} for retention at $k = 16$. For `bn2_w16_gap` at $k = 8$, residual energy is 0.190 under release and 0.339 under retention.

Table S8 Observed and reconstructed differences in B between the structured and pixel-permuted banks, averaged across architectures on `two_concepts`. The final column is the paired observed-minus-additive comparison over 100 renderer blocks.

Metric	Observed	Additive	Paired difference [95% CI]
Centered-logit fidelity	+0.05743	+0.05710	+0.00033 [−0.00074, +0.00140]
Probability-error reduction	+0.04563	+0.04105	+0.00458 [+0.00202, +0.00714]

The sum of individual logit changes closely reproduces the spatial contrast averaged across the studied architectures. Differences between observed and reconstructed contrasts remain in individual architectures.

S7 Reproducibility map

Table S9 identifies the six main-manuscript figures and their archived input groups. The same archive also contains the inputs for five supplementary figures. The publicly accessible, versioned reproducibility artifact is available at doi:10.5281/zenodo.23432463. Its `manifest.json` records SHA-256 checksums and original-to-exported source paths; `reproduce.py` regenerates all eleven figure PDFs from the archived CSV/NPZ inputs without requiring the original repository or model checkpoints.

Table S9 Manuscript figures and compact reproducibility input groups. Each figure generator is supplied under `code/figures/`; the archival source filename for every input is recorded in `manifest.json`.

Fig.	Figure generator	Archive input group
1	Overview	Procedural renderer and fixed template bank; no CSV input
2	Compact filter gallery	Filter structure: matched kernel arrays
3	Direct additivity curves	Additivity: architecture comparison
4	Intervention-size and architecture results	Metric sensitivity; architecture comparison
5	Spatial filter-bank specificity	Spatial specificity
6	Observed and additive reconstruction	Additivity: architecture comparison and spatial controls

The BN2-GAP interval in Section S6 is independently recomputed from the archived 100-block table by the reproduction entry point. The spatial observed-minus-additive tables and their published inferential summaries are also supplied, but the compact runner does not repeat their original bootstrap calculations. Frozen and post-hoc protocols remain separately labeled in the archive's `protocols/` directory.

S8 Integrity checks and artifact scope

Full-patch and no-op identity checks are recorded in the archived evaluator summaries. For the analytic Tiny controls, observed and reconstructed centered-logit curves agree to floating-point precision. An independent checkpoint reimplementation audit of the earlier causal evaluator is included in the archive's checkpoint-audit data group.

The versioned reproducibility archive at doi:10.5281/zenodo.23432463 excludes raw training checkpoints and per-model evaluator JSON files, which are retained separately because of their size. Its derived tables and compact kernel arrays allow regeneration

of all eleven figure PDFs (six main and five supplementary) and selected independent numerical checks. Other reported inferential summaries are archived rather than recomputed by this compact entry point. The archive README and manifest document file provenance, checksums, and the reproduction boundary.